

Attention

A small English-to-French transformer, built the way Vaswani et al. described it in 2017: multi-head attention, sinusoidal positions, and a position-wise feed-forward network.

1,008,000 parameters · 2 layers · d_model 128 · 4 heads · 387 training sentences · validation token accuracy 67.8%

TranslateGo.Run!Hello.Thank you.How are you?I love you.Wait!Help!Good morning.He is a doctor.This is a book.

French

Je t'aime !

Read as i love you .

Steps

These figures are measured on the sentence you just translated.

1. Token ids

Token ids from the English vocabulary: i → 10, love → 200, you → 17, . → 4.

token	id	in vocabulary
i	10	yes
love	200	yes
you	17	yes
.	4	yes

2. Embedding plus position

Each id is a 128-dimensional vector multiplied by $\sqrt{128} = 11.314$. A sinusoid of the same width is then added. The table shows those lengths and the first four coordinates. Under it, the curves and the grid use the same dimensions.

If the love embedding is

[-0.0864, 0.0873, -0.1197, ...]

then after scaling, the first coordinates are

$$[-0.0864 \times 11.314, 0.0873 \times 11.314, -0.1197 \times 11.314, \dots]$$

which is about

$$[-0.977, 0.987, -1.354, \dots]$$

The table value $\|\text{word} \times \sqrt{d}\| = 10.798$ is the length of all 128 coordinates:

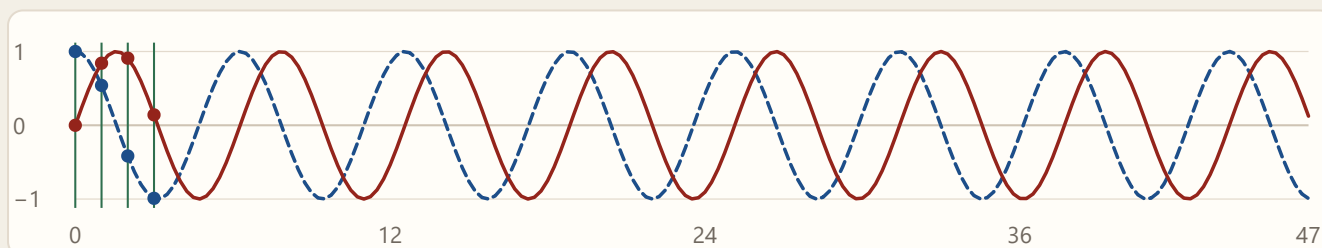
$$\sqrt{(x_1^2 + x_2^2 + \dots + x_{128}^2)} = \sqrt{116.603} = 10.798$$

token	$\ \text{word} \times \sqrt{d}\ $	$\ \text{position}\ $	$\ \text{sum}\ $	position dims 0–3	sum dims 0–3
i	10.059	8.000	11.646	0.000, 1.000, 0.000, 1.000	-0.407, -0.387, -0.698, 0.3
love	10.798	8.000	11.442	0.841, 0.540, 0.762, 0.648	-0.136, 1.528, -0.593, -0.5
you	9.793	8.000	12.216	0.909, -0.416, 0.987, -0.160	1.222, -0.955, -0.151, -1.19
.	8.677	8.000	10.054	0.141, -0.990, 0.517, -0.856	0.896, -1.327, 0.758, -0.01

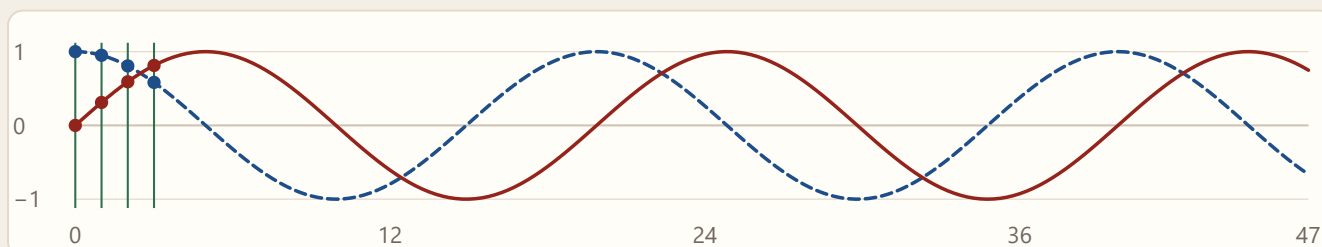
— sine, even dimension - - - cosine, odd dimension — this sentence

$\text{PE}(\text{position}, 2i) = \sin(\text{position} / 10000^{(2i/128)})$, and dimension $2i+1$ is the cosine. Dimension 0 repeats every 6.28 positions; dimension 64 repeats every 628.32. Ticks mark i at position 0, love at position 1, you at position 2, . at position 3.

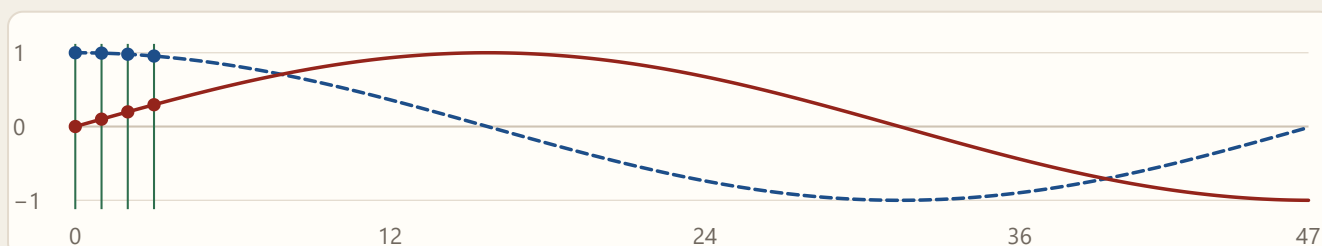
Dimensions 0 and 1. Repeats every 6.28 positions.



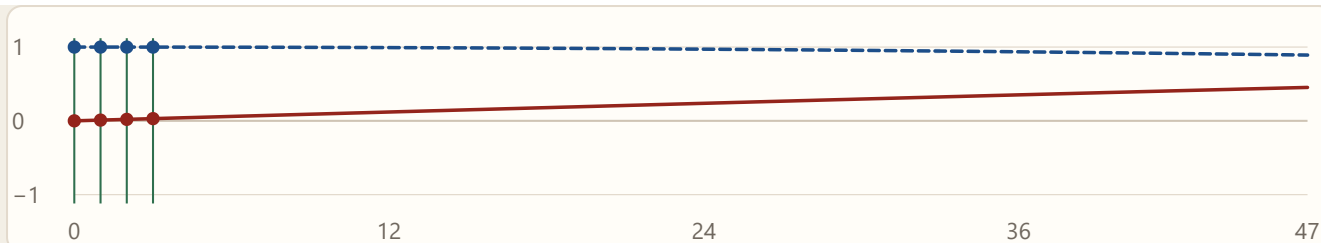
Dimensions 16 and 17. Repeats every 19.87 positions.



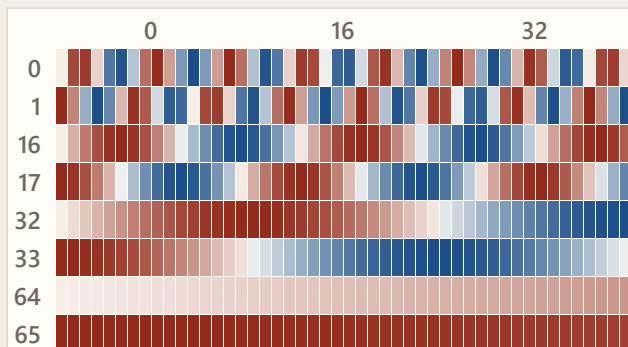
Dimensions 32 and 33. Repeats every 62.83 positions.



Dimensions 64 and 65. Repeats every 628.32 positions.



Dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the curves, at positions 0–47. Even rows are sines and odd rows are cosines.



Rows are dimensions 0, 1, 16, 17, 32, 33, 64, 65. Columns are positions 0–47. Even rows are sine and odd rows are cosine. Color is scaled to the largest absolute value here, 1.000.

Scaled word vectors at those dimensions

	0	1	16	17	32	33	64	65
i	-0.41	-1.39	-1.04	-1.35	0.43	-1.13	0.36	0.92
love	-0.98	0.99	-0.28	-1.66	-0.23	-0.61	-0.64	-0.72
you	0.31	-0.54	1.21	-2.21	-1.13	0.67	0.54	-1.74
.	0.76	-0.34	0.99	-1.22	0.03	0.13	0.90	-1.02

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.208.

Positional vectors at those dimensions

	0	1	16	17	32	33	64	65
i	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
love	0.84	0.54	0.31	0.95	0.10	0.99	0.01	1.00
you	0.91	-0.42	0.59	0.81	0.20	0.98	0.02	1.00
.	0.14	-0.99	0.81	0.58	0.30	0.95	0.03	1.00

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 1.000.

Sum that enters the first encoder layer, same dimensions

	0	1	16	17	32	33	64	65
i	-0.41	-0.39	-1.04	-0.35	0.43	-0.13	0.36	1.92
love	-0.14	1.53	0.03	-0.71	-0.13	0.38	-0.63	0.28
you	1.22	-0.95	1.80	-1.40	-0.93	1.65	0.56	-0.74
.	0.90	-1.33	1.80	-0.64	0.33	1.09	0.93	-0.02

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 1.916.

3. Encoder layers

Last of 2 encoder layers, mean of 4 heads. i attends most to i (0.5956) love attends most to love (0.7990) you attends most to love (0.9655) . attends most to love (0.9957). Both layers follow, with every head, the ReLU hidden state, and the residual stream.

↓ 2 encoder layers

└	Linear: XW_Q , XW_K , XW_V	← MatMul
└	QK^T	← MatMul
└	softmax	
└	$\text{attention_weights} \cdot V$	← MatMul
└	concat + output Linear	← MatMul
└	FFN 128 → 512	← MatMul
└	FFN 512 → 128	← MatMul

Each of the 2 encoder layers does this once. The blocks below are those tensors for this sentence. Wide products are sampled at dimensions 0, 1, 16, 17, 32, 33, 64, 65. QK^T and the softmax are the full token-by-token matrices. QK^T is divided by $\sqrt{32}$ before the softmax, and ReLU sits between the two feed-forward linears.

Encoder layer 1

Q from the query linear

	0	1	16	17	32	33	64	65
i	-2.06	-0.96	-1.08	-2.20	-0.62	2.42	0.29	0.78
love	-1.23	-0.79	1.37	-3.86	-0.74	-1.08	1.29	0.65
you	-1.84	-0.11	-0.06	-0.18	-0.32	0.19	-0.01	-0.43
.	-0.25	0.90	-0.45	2.70	-1.10	1.57	-0.99	0.16

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 3.859.

K from the key linear

	0	1	16	17	32	33	64	65
i	-0.31	-1.61	-0.07	-4.66	0.17	0.05	1.00	0.04
love	-0.55	0.95	1.08	0.96	1.73	-0.54	-0.06	-1.31
you	0.91	-1.50	-1.15	1.76	-0.30	-0.66	0.41	-2.48
.	1.71	0.11	-0.63	0.73	0.05	-1.67	-0.21	-1.22

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 4.656.

V from the value linear

	0	1	16	17	32	33	64	65
i	0.87	-1.42	0.59	1.64	0.81	0.14	0.14	-0.73

love	0.22	-0.01	-1.28	-0.15	-1.27	-0.21	-0.84	0.85
you	0.85	-0.84	-0.21	-0.98	0.10	-2.02	-0.15	-0.09
.	-0.17	0.84	1.16	-1.38	0.24	-0.26	-0.49	0.08

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.022.

QK^T for one head. Entries are the raw dot products, before dividing by $\sqrt{32}$.

head 1 ▾

	i	love	you	.
i	55.159	9.143	-11.319	-23.093
love	49.105	2.548	-16.207	-14.792
you	36.637	18.574	-15.185	-23.505
.	-15.190	15.108	-4.114	-13.766

Head 1, the i row, against i, love, you, ..

$$QK^T = [55.1593, 9.1431, -11.3194, -23.0931]$$

$$QK^T / \sqrt{32} = [9.7509, 1.6163, -2.0010, -4.0823]$$

$$\text{softmax} = [0.9997, 0.0003, 0.0000, 0.0000]$$

softmax of those scores. This is the self-attention weight.

head 1 ▾

	i	love	you	.
i	1.000	0.000	0.000	0.000
love	1.000	0.000	0.000	0.000
you	0.961	0.039	0.000	0.000
.	0.004	0.958	0.032	0.006

attention_weights · V, concatenated from 4 heads of 32

	0	1	16	17	32	33	64	65
i	0.87	-1.42	0.58	1.64	0.14	-1.15	-0.55	0.22
love	0.87	-1.42	0.58	1.64	0.11	-1.85	-0.83	0.84
you	0.85	-1.36	0.51	1.57	0.06	-1.91	-0.76	0.73
.	0.24	-0.04	-1.22	-0.18	-0.81	-0.68	-0.84	0.84

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 1.911.

concat + output linear

	0	1	16	17	32	33	64	65
i	0.89	0.19	-2.23	-1.31	-0.28	-1.02	0.92	-0.04
love	-0.02	1.35	-2.09	-1.02	-0.92	2.03	0.06	0.11
you	0.13	1.19	-1.46	-1.38	-0.63	2.14	0.43	-0.48
.	-0.63	0.39	-0.26	-1.68	0.49	2.20	0.91	-1.08

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.232.

FFN 128 → 512 before ReLU. Positive units: i 95/512, love 89/512, you 88/512, . 101/512.

	0	1	16	17	32	33	64	65
i	-0.46	-1.42	-0.41	1.24	2.44	-2.13	-0.40	-1.85
love	-1.04	-1.12	-0.59	1.33	1.52	-0.86	-0.64	-2.55
you	-1.03	-0.90	-0.35	1.38	0.79	-1.93	-0.55	-1.47
.	-1.15	-0.78	1.36	0.95	0.75	-1.78	-1.11	-1.96

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.554.

FFN 512 → 128, the second linear applied to the ReLU

	0	1	16	17	32	33	64	65
i	-0.54	-0.54	-0.17	-2.14	0.59	-1.09	3.48	1.03
love	-0.18	0.01	-0.15	-0.15	-0.33	1.69	-0.01	-0.70
you	-0.03	0.56	-0.26	-0.70	-0.22	0.84	-0.39	-1.29
.	-0.60	-0.83	0.61	0.04	0.89	0.90	-1.35	-1.80

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 3.484.

Encoder layer 2

Q from the query linear

	0	1	16	17	32	33	64	65
i	0.02	0.09	2.92	-0.61	0.45	-1.39	-0.92	-0.08
love	-0.55	-1.20	4.24	1.08	2.00	0.55	-0.03	-0.47
you	-0.65	-0.28	4.45	1.12	2.80	0.76	-0.38	0.61
.	-1.66	1.25	5.59	0.68	2.51	3.18	-0.91	0.31

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 5.586.

K from the key linear

	0	1	16	17	32	33	64	65
--	---	---	----	----	----	----	----	----

i	-0.20	-1.40	-1.12	-0.71	0.26	0.92	-0.13	0.38
love	-0.24	0.68	3.18	-0.31	2.75	2.28	0.10	0.76
you	0.00	-0.93	0.65	-0.78	2.46	0.51	0.89	-0.01
.	-0.43	-2.23	-0.65	-2.56	1.07	-1.58	-0.14	0.37

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 3.181.

V from the value linear

	0	1	16	17	32	33	64	65
i	0.36	-0.07	-0.04	-0.41	-0.64	-0.51	0.20	-0.11
love	-1.24	-0.99	0.53	-2.46	0.64	0.95	-0.43	-0.26
you	-0.57	0.80	0.20	-0.79	-0.28	1.65	-0.30	-0.62
.	-0.63	-0.78	0.76	-0.07	1.19	0.08	-1.97	-1.73

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.464.

QK^T for one head. Entries are the raw dot products, before dividing by $\sqrt{32}$.

head 1 ▾

	i	love	you	.
i	-13.497	23.465	-8.763	-25.328
love	-22.650	45.239	9.133	-1.021
you	-39.388	62.697	4.955	-5.869
.	-55.584	87.349	0.163	-26.295

Head 1, the i row, against i, love, you, ..

$$QK^T = [\text{ -13.4966, 23.4650, -8.7629, -25.3277 }]$$

$$QK^T / \sqrt{32} = [\text{ -2.3859, 4.1481, -1.5491, -4.4773 }]$$

$$\text{softmax} = [\text{ 0.0014, 0.9950, 0.0033, 0.0002 }]$$

softmax of those scores. This is the self-attention weight.

head 1 ▾

	i	love	you	.
i	0.001	0.995	0.003	0.000
love	0.000	0.998	0.002	0.000

you	0.000	1.000	0.000	0.000
.	0.000	1.000	0.000	0.000

attention_weights · V, concatenated from 4 heads of 32

	0	1	16	17	32	33	64	65
i	-1.24	-0.99	0.52	-2.46	-0.11	0.49	0.20	-0.12
love	-1.24	-0.99	0.52	-2.46	0.63	0.96	-0.70	-0.72
you	-1.24	-0.99	0.53	-2.46	0.62	0.96	-0.40	-0.26
.	-1.24	-0.99	0.53	-2.46	0.64	0.95	-0.43	-0.26

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.464.

concat + output linear

	0	1	16	17	32	33	64	65
i	-0.33	-1.17	-0.73	-0.99	-1.50	-1.07	1.48	-0.50
love	-0.34	0.74	1.44	-2.04	-1.24	0.35	-0.35	-0.85
you	-1.11	0.79	1.15	-2.28	-1.47	0.33	-0.44	-0.92
.	-1.17	0.87	1.22	-2.32	-1.44	0.39	-0.45	-0.96

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.319.

FFN 128 → 512 before ReLU. Positive units: i 233/512, love 189/512, you 182/512, . 150/512.

	0	1	16	17	32	33	64	65
i	-1.19	0.73	0.72	0.97	-1.09	2.17	-0.16	0.86
love	-0.35	0.85	-0.39	-0.40	-1.77	0.29	0.67	-0.35
you	-0.89	0.49	-0.14	-0.09	-1.46	0.33	0.48	-0.43
.	-0.31	0.62	-0.77	0.15	-1.42	-0.07	-0.03	-0.80

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.173.

FFN 512 → 128, the second linear applied to the ReLU

	0	1	16	17	32	33	64	65
i	-5.46	1.21	-2.62	-2.91	-0.40	-7.63	1.70	2.96
love	-3.81	2.61	-0.49	-3.97	-0.64	-4.81	-0.19	-0.14
you	-4.04	2.66	-1.68	-4.47	-0.55	-4.63	-0.14	-0.16
.	-2.73	1.54	-0.62	-2.96	-0.01	-3.74	-0.03	0.18

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 7.631.

Encoder layer 1

At i, 95 of 512 ReLU units are on. Residual length 11.422 after attention and 11.448 after the feed-forward block.

Self-attention. Rows and columns are the English tokens.

all heads, mean ▾

	i	love	you	.
i	0.329	0.078	0.147	0.447
love	0.287	0.261	0.451	0.001
you	0.258	0.357	0.359	0.026
.	0.009	0.891	0.085	0.015

Residual stream after attention

	0	1	16	17	32	33	64	65
i	0.17	-0.33	-2.44	-1.29	-0.07	-0.95	0.65	0.99
love	-0.23	1.54	-1.43	-1.18	-0.80	1.24	-0.50	0.03
you	0.67	-0.12	-0.09	-2.12	-1.36	2.23	0.37	-1.07
.	-0.07	-1.05	0.86	-2.11	0.37	2.23	1.10	-1.14

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.443.

Feed-forward ReLU, i 95/512, love 89/512, you 88/512, . 101/512 units on

	0	1	16	17	32	33	64	65
i	0.00	0.00	0.00	1.24	2.44	0.00	0.00	0.00
love	0.00	0.00	0.00	1.33	1.52	0.00	0.00	0.00
you	0.00	0.00	0.00	1.38	0.79	0.00	0.00	0.00
.	0.00	0.00	1.36	0.95	0.75	0.00	0.00	0.00

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.441.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
i	-0.15	-0.40	-1.28	-1.74	0.36	-1.01	2.23	1.09
love	-0.28	0.99	-0.99	-0.85	-0.73	1.87	-0.32	-0.46
you	0.49	0.34	-0.22	-2.02	-1.14	2.25	0.03	-1.70
.	-0.38	-1.16	0.98	-1.28	0.93	2.09	-0.09	-1.86

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.250.

Encoder layer 2

At i, 233 of 512 ReLU units are on. Residual length 11.374 after attention and 10.477 after the feed-forward block.

Self-attention. Rows and columns are the English tokens.

all heads, mean ▾

	i	love	you	.
i	0.596	0.253	0.100	0.051
love	0.074	0.799	0.051	0.075
you	0.023	0.966	0.010	0.002
.	0.002	0.996	0.002	0.000

Residual stream after attention

	0	1	16	17	32	33	64	65
i	-0.30	-0.87	-1.11	-1.51	-0.64	-1.18	1.98	0.28
love	-0.33	1.05	0.29	-1.63	-1.12	1.35	-0.36	-0.74
you	-0.30	0.67	0.53	-2.28	-1.39	1.46	-0.19	-1.41
.	-0.81	-0.10	1.21	-1.91	-0.21	1.42	-0.26	-1.53

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.277.

Feed-forward ReLU, i 233/512, love 189/512, you 182/512, . 150/512 units on

	0	1	16	17	32	33	64	65
i	0.00	0.73	0.72	0.97	0.00	2.17	0.00	0.86
love	0.00	0.85	0.00	0.00	0.00	0.29	0.67	0.00
you	0.00	0.49	0.00	0.00	0.00	0.33	0.48	0.00
.	0.00	0.62	0.00	0.15	0.00	0.00	0.00	0.00

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.173.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
i	-1.39	0.09	-0.93	-1.11	-0.26	-2.30	0.99	0.90
love	-1.24	1.22	-0.02	-1.77	-0.57	-1.11	-0.18	-0.26
you	-1.30	1.10	-0.33	-2.14	-0.63	-1.01	-0.11	-0.49
.	-1.26	0.56	0.27	-1.84	-0.08	-0.89	-0.12	-0.51

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.300.

The residual stream after the last encoder layer is the encoder output. Cross-attention in the decoder reads that sequence, and nothing else from the English side.

4. One decoder step

No French word has been chosen yet. This is the whole path from <bos> to the first probability.

4.1 Decoder input

<bos> is vocabulary id 1. Like an English token, it is looked up in the target embedding, multiplied by $\sqrt{128} = 11.314$, then added to position 0 of the sinusoid.

[-0.0011, 0.0218, -0.0919, ...]

then after scaling, the first coordinates are

[-0.0011 × 11.314, 0.0218 × 11.314, -0.0919 × 11.314, ...]

which is about

[-0.012, 0.247, -1.040, ...]

Position 0 of the sinusoid begins

[0.0000, 1.0000, 0.0000, 1.0000, ...]

Their sum is the 128-dimensional decoder input at this position. Its length is 10.425. The first four coordinates are

[-0.0125, 1.2469, -1.0399, 0.4672, ...]

Decoder input at <bos>, same dimensions as the sinusoid curves

	0	1	16	17	32	33	64	65
<bos>	-0.01	1.25	-0.35	0.85	0.14	1.31	-0.04	-0.57

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65. Color is scaled to the largest absolute value here, 1.307.

On the next iteration the prefix is two tokens, so the decoder receives two such vectors: $E(\text{<bos>}) + PE(0)$ and $E(\text{chosen word}) + PE(1)$. Section 5 is that growth. Section 6 shows every position of the finished prefix.

4.2 Decoder layer 1

At <bos>, 110 of 512 ReLU units are on. This row is only the <bos> position; the French token is not known yet.

Masked self-attention. The axes are the tokens already fed to the decoder, starting with <bos>.

all heads, mean ▾

	<bos>
<bos>	1.000

Cross-attention onto the encoder output. Columns are the English tokens.

all heads, mean ▾

	i	love	you	.
<bos>	0.363	0.097	0.151	0.389

Residual stream after masked self-attention

	0	1	16	17	32	33	64	65
<bos>	-0.06	1.72	-1.27	-0.70	-0.98	0.15	0.76	-0.13

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 1.722.

Residual stream after cross-attention

	0	1	16	17	32	33	64	65
<bos>	0.36	0.74	-1.84	0.01	-0.67	-0.08	0.47	-0.47

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 1.838.

Feed-forward ReLU, <bos> 110/512 units on

	0	1	16	17	32	33	64	65
<bos>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 0.000.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
<bos>	-2.08	0.53	-1.37	-0.86	0.27	-0.53	1.02	1.39

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.083.

4.3 Decoder layer 2

At <bos>, 126 of 512 ReLU units are on. This row is only the <bos> position; the French token is not known yet.

Masked self-attention. The axes are the tokens already fed to the decoder, starting with <bos>.

all heads, mean ▾

	<bos>
<bos>	1.000

Cross-attention onto the encoder output. Columns are the English tokens.

all heads, mean ▾

	i	love	you	.
--	---	------	-----	---

<bos>

0.990

0.004

0.005

0.001

Residual stream after masked self-attention

	0	1	16	17	32	33	64	65
<bos>	-2.61	0.79	-1.42	-1.37	-0.30	-1.68	0.50	1.64

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.614.

Residual stream after cross-attention

	0	1	16	17	32	33	64	65
<bos>	-3.25	0.11	-0.89	-0.92	0.60	-2.30	0.53	0.97

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 3.248.

Feed-forward ReLU, <bos> 126/512 units on

	0	1	16	17	32	33	64	65
<bos>	0.00	1.28	0.19	0.00	0.81	0.00	0.00	1.29

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 1.292.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
<bos>	-1.25	1.15	-1.00	-0.23	1.75	-2.75	0.78	0.67

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.752.

4.4 Vocabulary output projection

Only here does a French token appear. The vector h is the residual at <bos> after the last decoder layer.

After the last decoder layer, the <bos> position is one vector h with 128 coordinates. Its length is 13.163. The first four coordinates are

[-1.2460, 1.1484, 0.3732, 0.0880, ...]

$$z = h W_{\text{out}}$$

The output matrix W_{out} has one row per French token and 128 columns. It is the French embedding matrix, and this model adds no bias. Each logit is the dot product of h with one row. For j_e , the first three products $h_k W_{kj}$ are $0.1897 + 0.1542 + 0.0088$, and the sum over all 128 coordinates is the logit 8.3588.

That projection emits 323 logits, one for every French token. The largest are:

token	logit z	probability
j_e	8.3588	0.9258

demande	2.1397	0.0018
t	1.8820	0.0014
aha	1.8690	0.0014
quelle	1.3789	0.0009
!	1.3149	0.0008
entrez	1.3045	0.0008
chacun	1.2903	0.0008

Softmax turns those logits into probabilities. For je:

$$P = \exp(z) / \sum \exp(z)$$

$$P(\text{je}) = \exp(8.3588) / 4609.5806 = 0.9258$$

Section 5 repeats this cycle. Before each choice, <pad> and <bos> are removed, and on the first step so is <eos>. The winning token is appended to the prefix.

5. Autoregressive generation

The first iteration is section 4. Each later iteration appends the chosen token and runs the decoder again. <bos> → je → t → ' → aime → ! → <eos>.

5.1 Prefix <bos>

The decoder reads <bos>. je has probability 0.9258 (92.58%). It is appended, so the next prefix is <bos> je.

candidate	probability	log probability
je	0.9258	-0.0771
demande	0.0018	-6.2962
t	0.0014	-6.5539
aha	0.0014	-6.5669
quelle	0.0009	-7.0570

Beam kept after this choice. The highlighted row is continued. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je	-0.0771	1.0000	-0.0771
demande	-6.2962	1.0000	-6.2962
t	-6.5539	1.0000	-6.5539
aha	-6.5669	1.0000	-6.5669

5.2 Prefix <bos> je

The decoder reads <bos> je. t has probability 0.9537 (95.37%). It is appended, so the next prefix is <bos> je t.

candidate	probability	log probability
t	0.9537	-0.0474
ceci	0.0017	-6.3948
compatis	0.0007	-7.2814
tomber	0.0007	-7.2847
imbécile	0.0006	-7.3686

Beam kept after this choice. The highlighted row is continued. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t	-0.1245	1.0969	-0.1135
je ceci	-6.4719	1.0969	-5.9002
t '	-6.6116	1.0969	-6.0275
aha !	-6.6463	1.0969	-6.0591

5.3 Prefix <bos> je t

The decoder reads <bos> je t. ' has probability 0.9521 (95.21%). It is appended, so the next prefix is <bos> je t '.

candidate	probability	log probability
'	0.9521	-0.0491
aime	0.0008	-7.0775
prudemment	0.0008	-7.1683
t	0.0008	-7.1691
reviens	0.0006	-7.4527

Beam kept after this choice. The highlighted row is continued. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t '	-0.1736	1.1884	-0.1461
t ' aime	-6.6684	1.1884	-5.6113
aha ! <eos>	-6.7138	1.1884	-5.6495
je t aime	-7.2020	1.1884	-6.0602

5.4 Prefix <bos> je t '

The decoder reads <bos> je t '. aime has probability 0.9408 (94.08%). It is appended, so the next prefix is <bos> je t ' aime.

candidate	probability	log probability
aime	0.9408	-0.0610
!	0.0024	-6.0404
'	0.0009	-7.0196
je	0.0006	-7.3447
suggestions	0.0006	-7.3471

Beam kept after this choice. The highlighted row is continued. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t ' aime	-0.2346	1.2754	-0.1840
je t ' !	-6.2140	1.2754	-4.8721
t ' aime !	-6.7366	1.2754	-5.2818
je t ' '	-7.1932	1.2754	-5.6398

5.5 Prefix <bos> je t ' aime

The decoder reads <bos> je t ' aime. ! has probability 0.9338 (93.38%). It is appended, so the next prefix is <bos> je t ' aime !.

candidate	probability	log probability
!	0.9338	-0.0685
aime	0.0008	-7.0844
?	0.0005	-7.6584
t	0.0005	-7.6852
je	0.0004	-7.7381

Beam kept after this choice. The highlighted row is continued. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t ' aime !	-0.3031	1.3587	-0.2231
je t ' ! <eos>	-6.2814	1.3587	-4.6232
t ' aime ! <eos>	-6.8039	1.3587	-5.0078
je t ' ' aime	-7.2544	1.3587	-5.3394

5.6 Prefix <bos> je t ' aime !

The decoder reads <bos> je t ' aime !. <eos> has probability 0.9342 (93.42%). It is appended, so the next prefix is <bos> je t ' aime ! <eos>.

candidate	probability	log probability
<eos>	0.9342	-0.0681
aime	0.0003	-7.9831
t	0.0003	-8.0663
'	0.0003	-8.1250
?	0.0003	-8.1334

Beam kept after this choice. The highlighted row is continued. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t ' aime ! <eos>	-0.3712	1.4386	-0.2580
je t ' ' aime !	-7.3246	1.4386	-5.0914
je t ' aime ! aime	-8.2862	1.4386	-5.7598
je t ' aime ! t	-8.3693	1.4386	-5.8176

6. Decoder layers on the chosen prefix

The beam kept <bos> je t ' aime !. These maps are a diagnostic pass of that whole prefix through decoder layer 1 and layer 2, after the tokens were already chosen.

Target embedding plus position at every prefix token

	0	1	16	17	32	33	64	65
<bos>	-0.01	1.25	-0.35	0.85	0.14	1.31	-0.04	-0.57
je	-0.88	2.06	-0.48	1.72	1.74	-0.41	0.73	1.11
t	-0.56	0.48	0.24	1.48	-0.43	1.10	-1.10	0.12
'	0.21	-0.73	1.78	0.42	0.05	1.92	-0.78	1.72
aime	-0.17	-2.06	-0.17	-0.38	0.86	-0.67	-0.92	-0.16
!	-0.60	-0.19	2.17	0.14	-0.52	1.72	1.17	1.27

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.169.

Decoder layer 1

While writing je, 110 of 512 ReLU units are on. Residual length 12.31 after masked self-attention, 11.484 after cross-attention, and 11.524 after the feed-forward block.

Masked self-attention. The axes are the tokens already fed to the decoder, starting with <bos>.

all heads, mean ▼

	<bos>	je	t	'	aime	!
<bos>	1.000	0.000	0.000	0.000	0.000	0.000
je	0.500	0.500	0.000	0.000	0.000	0.000
t	0.247	0.253	0.500	0.000	0.000	0.000
'	0.250	0.000	0.250	0.499	0.000	0.000
aime	0.250	0.000	0.251	0.234	0.265	0.000
!	0.000	0.000	0.000	0.002	0.000	0.998

Cross-attention onto the encoder output. Columns are the English tokens.

all heads, mean ▾

	i	love	you	.
je	0.363	0.097	0.151	0.389
t	0.215	0.112	0.200	0.473
'	0.163	0.225	0.256	0.356
aime	0.105	0.415	0.201	0.279
!	0.233	0.547	0.139	0.082
<eos>	0.375	0.159	0.188	0.278

Residual stream after masked self-attention

	0	1	16	17	32	33	64	65
je	-0.06	1.72	-1.27	-0.70	-0.98	0.15	0.76	-0.13
t	-0.47	0.80	-1.19	0.61	0.45	-0.88	0.88	0.47
'	-0.72	-0.57	0.28	0.57	0.47	0.37	-0.69	-0.23
aime	0.79	-1.44	1.02	-0.67	0.39	0.54	-0.35	-0.06
!	-0.56	-2.71	1.02	-2.04	0.88	0.18	-0.50	-0.26
<eos>	2.02	-0.64	0.93	0.43	-1.57	0.74	0.82	0.76

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.712.

Residual stream after cross-attention

	0	1	16	17	32	33	64	65
je	0.36	0.74	-1.84	0.01	-0.67	-0.08	0.47	-0.47
t	0.08	-0.08	-1.82	0.83	0.24	-0.97	0.65	-0.03
'	-0.00	-1.30	-0.67	0.68	0.23	-0.06	-0.62	-0.52

aime	1.17	-2.16	-0.21	-0.19	0.12	0.13	-0.13	-0.41
!	0.28	-2.96	-0.05	-1.41	0.60	0.00	-0.41	-0.37
<eos>	2.17	-1.29	-0.24	0.81	-1.16	0.50	0.65	0.18

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.961.

Feed-forward ReLU, je 110/512, t 126/512, ' 100/512, aime 137/512, ! 199/512, <eos> 236/512 units on

	0	1	16	17	32	33	64	65
je	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
t	0.00	0.00	0.00	0.00	2.40	0.00	0.00	0.00
'	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
aime	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.46
!	0.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<eos>	0.00	0.00	0.00	1.91	0.00	0.00	0.00	0.00

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.405.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
je	-2.08	0.53	-1.37	-0.86	0.27	-0.53	1.02	1.39
t	-1.55	-0.18	-1.35	0.26	-0.32	-0.48	0.79	1.22
'	-0.14	-1.08	0.49	-0.16	1.14	0.20	-0.71	1.60
aime	1.48	-1.47	-0.65	-0.25	-0.58	-1.36	-1.13	-1.02
!	1.01	-2.27	1.24	0.12	0.16	0.09	0.15	-0.07
<eos>	1.86	0.23	0.09	-0.51	-1.36	1.03	1.53	1.01

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.271.

Decoder layer 2

While writing je, 126 of 512 ReLU units are on. Residual length 11.468 after masked self-attention, 11.305 after cross-attention, and 13.163 after the feed-forward block.

Masked self-attention. The axes are the tokens already fed to the decoder, starting with <bos>.

all heads, mean ▾

	<bos>	je	t	'	aime	!
<bos>	1.000	0.000	0.000	0.000	0.000	0.000
je	0.000	1.000	0.000	0.000	0.000	0.000
t	0.000	0.000	1.000	0.000	0.000	0.000

'	0.000	0.000	0.001	0.999	0.000	0.000
aime	0.000	0.000	0.000	0.000	1.000	0.000
!	0.000	0.000	0.000	0.000	0.000	1.000

Cross-attention onto the encoder output. Columns are the English tokens.

all heads, mean ▾

	i	love	you	.
je	0.990	0.004	0.005	0.001
t	0.009	0.327	0.399	0.266
'	0.003	0.346	0.255	0.396
aime	0.285	0.212	0.176	0.327
!	0.013	0.816	0.112	0.060
<eos>	0.241	0.084	0.159	0.516

Residual stream after masked self-attention

	0	1	16	17	32	33	64	65
je	-2.61	0.79	-1.42	-1.37	-0.30	-1.68	0.50	1.64
t	-1.59	0.16	-1.28	-0.20	-0.72	-1.10	0.37	1.80
'	-0.25	-0.91	1.98	-0.10	1.60	0.71	-1.27	1.54
aime	0.69	-1.24	-0.92	-0.58	-0.61	-2.18	-0.81	0.42
!	0.93	-0.94	0.27	0.78	-0.47	0.49	0.98	-0.29
<eos>	2.21	0.07	-0.13	0.15	-0.65	0.54	1.43	0.12

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.614.

Residual stream after cross-attention

	0	1	16	17	32	33	64	65
je	-3.25	0.11	-0.89	-0.92	0.60	-2.30	0.53	0.97
t	-3.13	-0.41	-1.40	0.98	0.22	-1.71	0.43	1.48
'	-2.33	-1.17	0.81	1.10	1.75	-0.59	-0.67	1.34
aime	-1.29	-1.26	-0.86	0.72	0.43	-2.67	-0.52	0.69
!	-1.42	-1.03	-0.42	1.50	0.35	-0.54	0.85	0.27
<eos>	-0.43	-0.28	-0.36	0.99	0.28	-0.97	1.14	0.34

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 3.248.

Feed-forward ReLU, je 126/512, t 134/512, ' 167/512, aime 150/512, ! 211/512, <eos> 285/512 units on

	0	1	16	17	32	33	64	65
je	0.00	1.28	0.19	0.00	0.81	0.00	0.00	1.29
t	0.00	1.36	0.18	0.00	0.00	0.00	0.00	0.48
'	0.00	0.57	0.00	0.00	0.00	0.00	0.00	2.21
aime	0.00	1.02	1.23	0.00	0.17	0.00	0.00	0.29
!	0.00	0.41	0.00	0.20	0.66	0.00	0.00	0.26
<eos>	0.00	0.49	0.97	0.00	0.00	0.31	0.86	1.53

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.210.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
je	-1.25	1.15	-1.00	-0.23	1.75	-2.75	0.78	0.67
t	-1.91	0.16	-0.03	0.11	-0.33	-1.18	-0.39	0.21
'	0.27	-0.71	3.00	-0.63	0.40	0.79	-1.03	0.76
aime	0.68	-1.35	0.44	-0.67	-0.15	-1.29	-0.13	-0.72
!	0.27	-1.14	1.27	-0.99	-1.60	0.63	1.57	0.68
<eos>	2.35	0.64	1.55	-0.00	0.46	2.00	0.20	-1.57

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.995.

7. Join the tokens

je t ' aime ! → Je t'aime !

Cross-attention

Each row is a French token. The color is how strongly the last decoder layer looked at each English token while writing it. Color is scaled within the row; the tooltip is the real weight.

Head all heads, mean ▼

	I	love	you	.
je				
t				
'				
aime				
!				

$\text{softmax}(Q K^T / \sqrt{d_k}) V$

Training sentences are short pairs from the Tatoeba Project, compiled at manythings.org (file dated 2026-02-13) under CC BY 2.0. Per-sentence credits are in `data/fra.txt`. The full paper model is much larger and was trained on WMT; this one is a CPU-sized copy of the architecture.